Math 110 Review

Quiz: minimal polynomial and generalized eigenvectors (Notes Ch. 13–14, LADR 5B, 8A, 8B)

§1 Review

§1.1 Tools from earlier chapters

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Every proof below uses only these facts. Each has its reason, except the fundamental theorem of algebra (it needs analysis).

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Why.

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: T of any vector is a combination of the Tu_i (which are 0) and the Tv_j.
3. Independent: if $\sum c_j Tv_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, and independence of the basis of V gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

Why. $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of n vectors is a basis.

(b) A subspace of dimension n is all of V.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Why.

1. (a), (b): an independent list extends to a basis, and every basis has length n.
2. (c): a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis x_i, u_j of U and, separately, to a basis x_i, w_l of W. Together the x_i, u_j, w_l span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ (x, u, w combinations of the x_i, u_j, w_l), then $w = -x - u \in U \cap W$, so w is a combination of the x_i. The basis x_i, w_l of W is independent, so $w = 0$. Then $x + u = 0$, and independence of x_i, u_j gives $x = u = 0$.

(T4) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

Why. $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.

Why. Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda)=0 \iff (z-\lambda) \mid p$.

Why. Divide p by $z-\lambda$: the remainder is a constant, and evaluating at λ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) λ is an eigenvalue $\iff$ $T-\lambda I$ is not injective.

(b) If $Tv=\lambda v$, then $p(T)v=p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why.

1. (a) is the definition: $Tv=\lambda v$ with $v \neq 0$ means $(T-\lambda I)v=0$ with $v \neq 0$.
2. (b): $T^k v=\lambda^k v$ by induction; add up the terms of p.
3. (d): take $v \neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v=0$ for some $p \neq 0$; p is not constant since $v \neq 0$.
4. Factor $p=c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v \neq 0$ to 0, so some $T-\lambda_j I$ is not injective (T4).

(T8) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1, \ldots, v_n \iff$ every $\operatorname{span}(v_1, \ldots, v_j)$ is invariant. Then the eigenvalues of T are the diagonal entries. Over $\mathbb{C}$ every operator has an upper-triangular matrix.

Why.

1. Column j holds the coordinates of Tv_j, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1, \ldots, v_j)$ for every j.
2. If $a_{jj}=\lambda$: $T-\lambda I$ maps $\operatorname{span}(v_1, \ldots, v_j)$ into $\operatorname{span}(v_1, \ldots, v_{j-1})$, a smaller space, so it is not injective (T1). So λ is an eigenvalue.
3. If no a_{jj} equals λ: $\mathcal{M}(T-\lambda I)$ is triangular with nonzero diagonal. Its columns are independent: in a relation, the right-most column used is the only one with a nonzero entry in its own diagonal row. So $T-\lambda I$ is injective.
4. Existence over $\mathbb{C}$, by induction on n: take an eigenvalue λ (T7(d)). $U=\operatorname{range}(T-\lambda I)$ is invariant (T5) and $\dim U < n$, since $T-\lambda I$ is not injective (T1).
5. By induction U has a basis $u_1, \ldots, u_m$ with every $\operatorname{span}(u_1, \ldots, u_j)$ invariant. Extend it to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V.
6. $Tv_j=(T-\lambda I)v_j+\lambda v_j \in U+\operatorname{span}(v_j) \subseteq \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_j)$. So every span of an initial part of the list is invariant.

(T9) **Division with remainder.** For polynomials p and $s \neq 0$ there are unique q, r with $p=sq+r$ and $r=0$ or $\deg r < \deg s$.

Why.

1. Existence, by induction on $\deg p$: if $\deg p < \deg s$, take $q=0$, $r=p$. Otherwise subtract $\frac{a}{b}x^{\deg p-\deg s}s$ from p (a, b the leading coefficients of p, s); this lowers the degree, so induction applies.
2. Uniqueness: if $sq+r=sq'+r'$, then $s(q-q')=r'-r$. The right side has degree $< \deg s$, so $q=q'$ and $r=r'$.

(T10) **Block diagonal matrices.** If $V=U \oplus W$ with both T-invariant, then in a basis of U followed by one of W, $\mathcal{M}(T)=\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

Why. $Tu \in U$ is a combination of the U-basis only, so its column is 0 in the W-rows (same for W). If $T(u+w)=0$, then $Tu=-Tw \in U \cap W=\{0\}$.

(T11) **Change of basis.** If column j of S holds the B-coordinates of the j-th vector of B', then $\mathcal{M}_{B'}(T)=S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

Why. S converts B'-coordinates to B-coordinates: apply S, then $\mathcal{M}_B(T)$, then convert back with S^{-1}. Every invertible S arises this way, since its columns form a basis.

(T12) **Quotient operator.** If U is T-invariant, then $(T/U)(v+U)=Tv+U$ is a well-defined operator on V/U, $\dim V/U=\dim V-\dim U$, and $p(T/U)(v+U)=p(T)v+U$.

Why.

1. Well defined: if $v+U=v'+U$, then $v-v' \in U$, so $Tv-Tv' \in U$ (U is invariant).

2. Dimension: $v \mapsto v + U$ maps V onto V/U with null space U; apply (T1).
3. $(T/U)^k(v + U) = T^k v + U$ by induction; add up the terms of p.

§1.2 Minimal polynomial (Ch. 13)

The question. Powers of T cannot stay independent forever, so some nonzero polynomial kills T. The one of smallest degree tells you the eigenvalues of T and whether T is diagonalizable.

- $T = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$: $(T - 2I)(T - 3I) = 0$, and no polynomial of degree 1 kills T, since T is not a multiple of I. The roots $2, 3$ are the eigenvalues.
- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $N^2 = 0$ but $N \neq 0$, so the smallest polynomial is x^2. Its repeated root is what stops N from being diagonalizable.

Definition (Def. 35, 36 / LADR 5.14, 5.22, 5.24)**.** For $p(x) = a_m x^m + \cdots + a_1 x + a_0$, $p(T) = a_m T^m + \cdots + a_1 T + a_0 I$. Let k be the smallest index with $T^k \in \operatorname{span}(I, T, \ldots, T^{k-1})$, and write $T^k = a_{k-1}T^{k-1} + \cdots + a_0 I$. The **minimal polynomial** of T is $m_T(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_0$. (The notes call it p.)

- k exists: $\mathcal{L}(V)$ has dimension n^2 (operators are $n \times n$ matrices), so the $n^2 + 1$ operators $I, T, \ldots, T^{n^2}$ are dependent. Solving a dependence for the highest power with a nonzero coefficient puts that power in the span of the lower ones.
- $I, T, \ldots, T^{k-1}$ are independent, by the same argument: a dependence among them would put some T^j with $j < k$ in the span of lower powers. So the a_i are unique.
- Shorthand used below: q *kills* T means $q(T) = 0$; an operator S *kills* v means $Sv = 0$.
- m_T is monic and $m_T(T) = 0$. No nonzero polynomial of degree $< k$ kills T, since it would be a dependence among $I, \ldots, T^{k-1}$. So m_T is the unique monic polynomial of least degree that kills T (two of them would differ by a nonzero polynomial of lower degree that kills T).
- Ex. 57: $m_I = x - 1$. For $T = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, I, T are independent and $T^2 = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right) = 4T - 3I$, so $m_T = x^2 - 4x + 3 = (x - 1)(x - 3)$.

Proposition (Prop. 65 / LADR 5.29)
For $q \in \mathcal{P}(\mathbb{F})$: $q(T) = 0 \iff m_T \mid q$.

Where the proof comes from

Want. Every polynomial that kills T is a multiple of m_T.

Know. m_T has the smallest degree among nonzero polynomials that kill T. Polynomials can be divided with remainder, and the remainder has smaller degree than the divisor.

Key step. Divide q by m_T. The remainder kills T too, and its degree is too small for it to be nonzero.

Proof. 1. If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$ (T5).
2. If $q(T) = 0$, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$.
3. A nonzero s of degree $< \deg m_T$ cannot kill T. So $s = 0$ and $m_T \mid q$.

□

Proposition (Prop. 66 / LADR 5.27)
The eigenvalues of T are exactly the roots of m_T in $\mathbb{F}$. If $m_T = (x - \lambda)s$, then $s(T) \neq 0$, and every nonzero vector $s(T)w$ is an eigenvector for λ.

Where the proof comes from

Want ($\Rightarrow$). $m_T(\lambda) = 0$ for each eigenvalue λ.

Know. On an eigenvector, a polynomial in T acts as the number $p(\lambda)$ (T7). Apply this to m_T, which is 0 as an operator.

Want ($\Leftarrow$). An eigenvector for a root λ, i.e. a nonzero vector killed by $T - \lambda I$.

Key step. Factor $m_T = (x - \lambda)s$. Then $(T - \lambda I)s(T) = m_T(T) = 0$, so $T - \lambda I$ kills every vector of the form $s(T)w$. Such a vector can be chosen nonzero because s has smaller degree than m_T, so $s(T) \neq 0$.

Proof. 1. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$.
2. If $m_T(\lambda) = 0$, write $m_T = (x - \lambda)s$ (T6). Since $\deg s < \deg m_T$, $s(T) \neq 0$; pick w with $s(T)w \neq 0$.
3. $(T - \lambda I)\big(s(T)w\big) = m_T(T)w = 0$, so $s(T)w$ is an eigenvector for λ.

The notes argue the second half by contradiction: if $T - \lambda I$ were invertible, multiplying $0 = s(T)(T - \lambda I)$ by $(T - \lambda I)^{-1}$ would give $s(T) = 0$. □

Proposition (Prop. 67 / LADR 5.22)

$\deg m_T \leq n$.

Where the proof comes from

Want. Some polynomial of degree $\leq n$ that kills T. Then m_T divides it (Prop. 65).

Know. For one vector $v \neq 0$, the $n + 1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $q(T)v = 0$ for some $q \neq 0$ of degree $\leq n$. This kills v, but not necessarily all of V.

First try. Do this for each basis vector and multiply the q's. The product kills T, but its degree can be about n^2.

Key step. Induction on $\dim V$. $U = \operatorname{null} q(T)$ is invariant and contains v. If $U \neq V$, both U and V/U are smaller than V. By induction, a polynomial r of degree $\leq \dim U$ kills T on U, and a polynomial p of degree $\leq \dim V/U$ kills T/U, which means $p(T)$ sends V into U. So $r(T)p(T)$ kills V, and its degree is $\leq \dim U + \dim V/U = n$.

Proof. 1. If $\dim V = 1$, $T = cI$ and $m_T = x - c$. In general, fix $v \neq 0$ and $q \neq 0$ with $\deg q \leq n$ and $q(T)v = 0$. $U = \operatorname{null} q(T)$ is invariant (T5).
2. If $q(T) = 0$, then $m_T \mid q$, so $\deg m_T \leq n$.
3. Otherwise $0 < \dim U < n$. By induction, $r = m_{T|_U}$ has $\deg r \leq \dim U$ and $p = m_{T/U}$ has $\deg p \leq \dim V/U$ (T12).
4. For $v \in V$: $p(T)v + U = p(T/U)(v + U) = 0 + U$, so $p(T)v \in U$, so $r(T)p(T)v = 0$. Hence $m_T \mid rp$, and $\deg(rp) \leq \dim U + \dim V/U = n$ (T12).

□

Second proof (LADR 5.22), without quotient spaces. Same plan, with $\operatorname{range} q(T)$ in place of V/U. Induction on n.

1. Fix $v \neq 0$ and let q be the monic polynomial of least degree d with $q(T)v = 0$; $d \geq 1$ since $v \neq 0$.
2. $v, Tv, \ldots, T^{d-1}v$ are independent (a dependence would be a smaller polynomial killing v), and they lie in $\operatorname{null} q(T)$, because $q(T)T^j v = T^j q(T)v = 0$. So $\dim \operatorname{null} q(T) \geq d$.
3. $W = \operatorname{range} q(T)$ is invariant (T5) and $\dim W \leq n - d < n$ (T1). By induction some s of degree $\leq n - d$ kills $T|_W$.
4. For every u: $q(T)u \in W$, so $s(T)q(T)u = 0$. So sq kills T, and $\deg(sq) \leq n$.

- Ex. 58: every degree from 1 to n occurs.
  - $m_I = x - 1$ in every dimension.
  - The shift N ($Ne_1 = 0$, $Ne_j = e_{j-1}$) has $m_N = x^n$, since $N^{n-1}e_n = e_1 \neq 0$ and $N^n = 0$.
  - Degree k, $1 \leq k \leq n$: let $Ne_1 = 0$, $Ne_j = e_{j-1}$ for $2 \leq j \leq k$, $Ne_j = 0$ for $j > k$. Then $N^k = 0$ and $N^{k-1}e_k = e_1 \neq 0$, so $m_N = x^k$.

Proposition (Prop. 68 / LADR 5.62)
T is diagonalizable $\iff$ $m_T = (x-\lambda_1)\cdots(x-\lambda_m)$ with $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$ distinct.

Where the proof comes from

Want ($\Rightarrow$). $\prod_i (T - \lambda_i I) = 0$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues.

Key step. On an eigenvector for λ_j, the factor $T - \lambda_j I$ gives 0, and the factors commute, so let that factor act first. There is an eigenbasis, so the product kills V. Then m_T divides $\prod(x - \lambda_i)$ and contains every $x - \lambda_i$ (Prop. 66), so it equals it.

Want ($\Leftarrow$). Write every v as a sum of eigenvectors.

Know. $(T - \lambda_j I)\prod_{i \neq j}(T - \lambda_i I) = m_T(T) = 0$. So every vector of the form $\prod_{i \neq j}(T - \lambda_i I)w$ is in $E(\lambda_j, T)$. This is how to produce eigenvectors.

Key step. Scale these so that they add up to v. The polynomial $g_j(x) = \prod_{i \neq j} \frac{x - \lambda_i}{\lambda_j - \lambda_i}$ is 1 at λ_j and 0 at the other λ_i. Then $\sum_j g_j - 1$ has degree $< m$ and m roots, so $\sum_j g_j = 1$. Apply both sides to v.

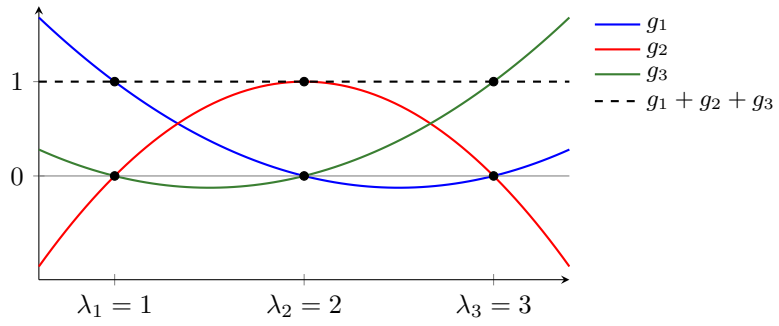
Proof. 1. ($\Rightarrow$) Let $v_1, \ldots, v_n$ be an eigenbasis, $Tv_k = \mu v_k$ with μ one of the λ_i. Letting the factor $T - \mu I$ act first, $\prod_i (T - \lambda_i I)v_k = 0$ for every k (T5). So $m_T \mid \prod_i (x - \lambda_i)$ (Prop. 65). Each λ_i is a root of m_T (Prop. 66), so $m_T = \prod_i (x - \lambda_i)$.

2. ($\Leftarrow$) Define g_j as above. $\sum_j g_j - 1$ has degree $\leq m - 1$ and vanishes at the m points $\lambda_1, \ldots, \lambda_m$. A nonzero polynomial of degree $\leq m - 1$ has at most $m - 1$ roots (each root splits off a factor, T6), so $\sum_j g_j = 1$.

3. For $v \in V$: $v = \sum_j g_j(T)v$, and $(T - \lambda_j I)g_j(T)v = c_j\, m_T(T)v = 0$, where $c_j = \prod_{i \neq j}(\lambda_j - \lambda_i)^{-1}$. So $g_j(T)v \in E(\lambda_j, T)$.

4. So $V = E(\lambda_1, T) + \cdots + E(\lambda_m, T)$. This sum is direct (T7), so bases of the eigenspaces together form an eigenbasis.

□



g_j is 1 at λ_j and 0 at the other eigenvalues, and $g_1 + g_2 + g_3 = 1$. Applying $1 = \sum g_j$ to T writes $v = \sum g_j(T)v$, and each $g_j(T)v$ lies in $E(\lambda_j, T)$ (Prop. 68). The same $g_j(T)$ are the projections of F9.

Second proof of ($\Leftarrow$), the one in the notes. Instead of splitting v into all its eigenvector pieces at once, split off the λ_1-piece and use induction for the rest. Let $g = \prod_{i \geq 2}(x - \lambda_i)$, so $m_T = (x - \lambda_1)g$ and $g(\lambda_1) \neq 0$.

1. Dividing by $x - \lambda_1$ leaves the remainder $g(\lambda_1)$ (T6): $g = (x - \lambda_1)q + g(\lambda_1)$.
2. For $v \in V$ let $u = g(T)v/g(\lambda_1)$. Then $(T - \lambda_1 I)u = m_T(T)v/g(\lambda_1) = 0$, so $u \in \text{null}(T - \lambda_1 I)$, and $v - u = -(T - \lambda_1 I)q(T)v/g(\lambda_1) \in \text{range}(T - \lambda_1 I)$.
3. So $V = \text{null}(T - \lambda_1 I) + \text{range}(T - \lambda_1 I)$. The two dimensions add up to n (T1), so the intersection is 0 (last item of T3) and the sum is direct.
4. $\text{range}(T - \lambda_1 I)$ is invariant, smaller than V (since $\text{null}(T - \lambda_1 I) \neq 0$: λ_1 is an eigenvalue by Prop. 66), and its minimal polynomial divides m_T (F3), so it again has distinct linear factors. By induction T restricted to it has an eigenbasis; add a basis of $E(\lambda_1, T)$.

More facts about m_T

Throughout, write $m_T = x^d + c_{d-1}x^{d-1} + \cdots + c_1 x + c_0$.

Proposition (F1: invertibility, HW7 P3)

T is invertible $\iff$ $c_0 \neq 0$. In that case $T^{-1} = -\dfrac{1}{c_0}\big(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\big)$, a polynomial in T.

Where the proof comes from

Want. T^{-1} written with powers of T.
Know. $m_T(T) = 0$, and every term of $m_T(T)$ except $c_0 I$ contains a factor T.
Key step. Move $c_0 I$ to the other side and factor out T. That gives $T \cdot (\text{something}) = -c_0 I$, so $\frac{-1}{c_0}(\text{something})$ is the inverse.

Proof. 1. $c_0 = m_T(0)$. By Prop. 66, $c_0 = 0$ $\iff$ 0 is an eigenvalue $\iff$ T is not injective (T7) $\iff$ T is not invertible (T2).
2. If $c_0 \neq 0$: $m_T(T) = 0$ says $T^d + c_{d-1}T^{d-1} + \cdots + c_1 T = -c_0 I$.
3. Factor T out of the left side: $T\big(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\big) = -c_0 I$. Divide by $-c_0$.

□

Proposition (F2: degree one, M4)

If $V \neq \{0\}$: $\deg m_T = 1$ $\iff$ $T = cI$ for some $c \in \mathbb{F}$.

Proof. 1. ($\Rightarrow$) $m_T = x - c$ and $m_T(T) = 0$ say $T - cI = 0$.
2. ($\Leftarrow$) $x - c$ kills cI. No polynomial of degree 0 does: a nonzero constant a gives $aI \neq 0$ since $V \neq \{0\}$. So $m_{cI} = x - c$.

□

Proposition (F3: restrictions and quotients, Ex. 60)

Let U be T-invariant.

(a) $m_{T|_U} \mid m_T$ and $m_{T/U} \mid m_T$.
(b) A monic divisor of a product $(x - \lambda_1)\cdots(x - \lambda_m)$ of distinct linear factors is the product of some of those factors.
(c) So if T is diagonalizable, so are $T|_U$ and T/U.

Proof. (a)

1. For $u \in U$: $m_T(T|_U)u = m_T(T)u = 0$. So m_T kills $T|_U$, and $m_{T|_U} \mid m_T$ by Prop. 65 applied to $T|_U$.
2. For $v + U \in V/U$: $m_T(T/U)(v + U) = m_T(T)v + U = 0 + U$ (T12). So $m_{T/U} \mid m_T$.

(b) Let s be the monic divisor, $st = \prod_{i=1}^{m}(x - \lambda_i)$.

1. Each λ_i makes the product 0, so it is a root of s or of t. Let S be the set of λ_i that are roots of s, and R the remaining λ_i (roots of t).
2. Distinct roots split off one factor each (T6), so $\prod_{\lambda\in S}(x-\lambda)$ divides s and $\prod_{\lambda\in R}(x-\lambda)$ divides t. Hence $\deg s \geq |S|$ and $\deg t \geq |R|$.
3. But $\deg s + \deg t = m = |S| + |R|$, so both are equalities. A monic s divisible by a monic polynomial of the same degree equals it: $s = \prod_{\lambda\in S}(x - \lambda)$.

(c) m_T is a product of distinct linear factors (Prop. 68). By (a) and (b), so are $m_{T|_U}$ and $m_{T/U}$. Apply Prop. 68 again. □

Proposition (F4: direct sums, block diagonal matrices)

If $V = U_1 \oplus \cdots \oplus U_r$ with every U_i invariant, then $m_T = \operatorname{lcm}\big(m_{T|_{U_1}}, \ldots, m_{T|_{U_r}}\big)$. For a block diagonal matrix, m is the lcm of the minimal polynomials of the blocks.

Proof. 1. Write $v = u_1 + \cdots + u_r$ with $u_i \in U_i$. Then $q(T)v = \sum_i q(T|_{U_i})u_i$, so $q(T) = 0 \iff q(T|_{U_i}) = 0$ for every i.
2. By Prop. 65, that holds $\iff$ every $m_{T|_{U_i}}$ divides q.
3. So m_T is the monic polynomial of least degree divisible by all the $m_{T|_{U_i}}$: their lcm. The blocks of the matrix are the matrices of the $T|_{U_i}$ (T10).

□

Proposition (F5: similarity, transpose, dual)

Similar matrices have the same minimal polynomial. Also $m_{A^\intercal} = m_A$ and $m_{T'} = m_T$.

Proof. 1. $(S^{-1}AS)^k = S^{-1}A^kS$ (the inner factors SS^{-1} cancel), so $p(S^{-1}AS) = S^{-1}p(A)S$. This is 0 exactly when $p(A) = 0$. So the same polynomials kill both, and the minimal polynomials agree.
2. $(A^\intercal)^k = (A^k)^\intercal$, so $p(A^\intercal) = p(A)^\intercal$, which is 0 exactly when $p(A) = 0$.
3. In dual bases, $\mathcal{M}(T') = \mathcal{M}(T)^\intercal$; apply step 2.

□

Proposition (F6: shifts, scalings, inverses, HW7 P4, M5)

With $d = \deg m_T$:

(a) $m_{T-\lambda I}(x) = m_T(x + \lambda)$;
(b) $m_{cT}(x) = c^d\, m_T(x/c)$ for $c \neq 0$;
(c) if T is invertible, $m_{T^{-1}}(x) = \frac{1}{c_0}x^d m_T(1/x)$, i.e. reverse the list of coefficients of m_T and divide by c_0.

Where the proof comes from

Want. The minimal polynomial of a new operator built from T.

Know. A polynomial in the new operator is a polynomial in T after a substitution: $q(T - \lambda I)$ is $r(T)$ with $r(x) = q(x - \lambda)$.

Key step. The substitution can be undone and does not change degrees. So it matches the polynomials that kill the new operator with the polynomials that kill T, and the one of least degree goes to the one of least degree.

Proof. (a)

1. For any q, $q(T - \lambda I) = r(T)$ with $r(x) = q(x - \lambda)$.
2. So q kills $T - \lambda I \iff m_T(x) \mid q(x - \lambda)$ (Prop. 65) $\iff m_T(x + \lambda) \mid q(x)$ (replace x by $x + \lambda$ throughout).
3. $m_T(x + \lambda)$ is monic, so it is $m_{T-\lambda I}$.

(b) The same three steps with $r(x) = q(cx)$: q kills $cT \iff m_T(x/c) \mid q(x)$. Multiply $m_T(x/c)$ by c^d to make it monic.

(c)

1. Let $r(x) = \frac{1}{c_0}x^d m_T(1/x) = \frac{1}{c_0}\big(c_0x^d + c_1x^{d-1} + \cdots + c_{d-1}x + 1\big)$. It is monic of degree d ($c_0 \neq 0$ by F1).
2. $r(T^{-1}) = \frac{1}{c_0}\big(c_0T^{-d} + c_1T^{-(d-1)} + \cdots + I\big) = \frac{1}{c_0}T^{-d}m_T(T) = 0$. So $m_{T^{-1}} \mid r$ and $\deg m_{T^{-1}} \leq d$.
3. The same argument with T^{-1} in place of T (its inverse is T) gives $d \leq \deg m_{T^{-1}}$.
4. So $m_{T^{-1}}$ divides r and has the same degree; both are monic, so they are equal.

□

Proposition (F7: one vector at a time)

For $v \in V$, let p_v be the monic polynomial of least degree with $p_v(T)v = 0$.

(a) $q(T)v = 0 \iff p_v \mid q$. In particular $p_v \mid m_T$.
(b) For a basis $v_1, \ldots, v_n$: $m_T = \operatorname{lcm}(p_{v_1}, \ldots, p_{v_n})$.
(c) If $v, Tv, \ldots, T^{n-1}v$ is a basis (v is a **cyclic vector**), then $m_T = p_v$, which has degree n.

Where the proof comes from

Want. m_T from computations with single vectors, which are cheaper than powers of a matrix.

Know. Polynomials that kill one vector behave like polynomials that kill T: the division argument of Prop. 65 works the same way. An operator that is 0 on a basis is 0.

Key step. Killing T means killing every basis vector, i.e. being divisible by every p_{v_i}. The least such polynomial is the lcm.

Proof. 1. (a) Divide: $q = p_v s + r$ with $\deg r < \deg p_v$ or $r = 0$. If $q(T)v = 0$, then $r(T)v = q(T)v - s(T)p_v(T)v = 0$, so $r = 0$ by minimality of p_v. Since $m_T(T)v = 0$, $p_v \mid m_T$.
2. (b) Let L be the lcm. Each $p_{v_i} \mid L$, so $L(T)v_i = 0$ by (a); $L(T)$ is 0 on a basis, so $L(T) = 0$ and $m_T \mid L$.
3. Each $p_{v_i} \mid m_T$ by (a), so $L \mid m_T$. Both are monic, so $L = m_T$.
4. (c) $v, \ldots, T^{n-1}v$ are independent, so no nonzero polynomial of degree $< n$ kills v: $\deg p_v \geq n$. Also $p_v \mid m_T$ and $\deg m_T \leq n$ (Prop. 67). So $m_T = p_v$.

□

Proposition (F8: real versus complex)

For a real matrix A, m_A is the same over $\mathbb{R}$ and over $\mathbb{C}$. So A is diagonalizable over $\mathbb{C} \iff m_A$ has distinct roots, and diagonalizable over $\mathbb{R} \iff$ in addition every root is real. (The notes prove this again in Ch. 23 by another method.)

Where the proof comes from

Want. Computing over $\mathbb{C}$ cannot give a different answer than over $\mathbb{R}$.

Key step. Def. 36 finds m_A by solving linear equations whose coefficients are entries of powers of A, all real. Solving linear equations never leaves the real numbers.

Proof. 1. For each k, "$A^k = a_{k-1}A^{k-1} + \cdots + a_0 I$" is a system of n^2 linear equations in $a_0, \ldots, a_{k-1}$ with real coefficients.
2. Row reduction uses only $+, -, \times, \div$ of real numbers, so the system has a complex solution iff it has a real one.
3. So the first solvable k is the same over $\mathbb{R}$ and $\mathbb{C}$, and the solution is unique (Def. 36), hence real. So m_A is the same.
4. The consequences are Prop. 68 over each field: over $\mathbb{R}$ the linear factors must have real roots.

□

Proposition (F9: projections onto eigenspaces, HW7 P7)

If $m_T = \prod_{i=1}^{m}(x - \lambda_i)$ with distinct roots, then $g_j(T)$, with g_j as in Prop. 68, is the projection onto $E(\lambda_j, T)$ along $\bigoplus_{i \neq j} E(\lambda_i, T)$.

Proof. On $E(\lambda_i, T)$, $g_j(T)$ acts as the number $g_j(\lambda_i)$ (T7), which is 1 for $i = j$ and 0 for $i \neq j$. Since $V = \bigoplus_i E(\lambda_i, T)$, $g_j(T)(v_1 + \cdots + v_m) = v_j$. □

Proposition (F10: repeated roots, Ex. 59) (a) λ is a repeated root of $p \iff p(\lambda) = p'(\lambda) = 0$.

(b) The common roots of p and p' are the roots of $\gcd(p, p')$, which the Euclidean algorithm finds without factoring p.

(c) Over $\mathbb{C}$, if $T^k = I$ for some $k \geq 1$, then T is diagonalizable.

Where the proof comes from

Want. A test for repeated roots that does not need the roots.

Know. From calculus: at a double root the graph touches the axis, so the value and the slope are both 0.

Key step. The product rule turns this into algebra: differentiating $(x-\lambda)^2 q$ leaves a factor $x - \lambda$.

Proof. (a)

1. If $p = (x-\lambda)^2 q$, then $p' = 2(x-\lambda)q + (x-\lambda)^2 q'$, which is 0 at λ.
2. Conversely, if $p(\lambda) = 0$, write $p = (x-\lambda)q$ (T6). Then $p' = q + (x-\lambda)q'$, so $p'(\lambda) = q(\lambda)$. If this is 0, then $(x-\lambda) \mid q$, so $(x-\lambda)^2 \mid p$.

(b) Divide p by p', then p' by the remainder, and so on. Each remainder is (previous dividend) $-$ (quotient)(previous divisor), so a common root of p, p' is a root of every remainder, in particular of the last nonzero one, $g = \gcd(p, p')$. Conversely, g divides p and p' (work back up the divisions).

Example: $p = x^3 - 3x + 2$, $p' = 3x^2 - 3$. Then $p = \frac{x}{3}\, p' + (-2x + 2)$, and $p' = -\frac{3}{2}(x+1) \cdot (-2x+2)$ with remainder 0. So g is a multiple of $x - 1$, and 1 is a double root: $p = (x-1)^2(x+2)$.

(c)

1. $T^k = I$ means $x^k - 1$ kills T, so $m_T \mid x^k - 1$.
2. $(x^k - 1)' = kx^{k-1}$ has only the root 0, which is not a root of $x^k - 1$. By (a), $x^k - 1$ has no repeated root, so it is a product of k distinct linear factors.
3. m_T is a monic divisor of it, so it is a product of distinct linear factors (F3(b)). Prop. 68.

□

Proposition (F11: odd dimension, LADR 5B)

Every operator on an odd-dimensional real vector space has an eigenvalue.

Where the proof comes from

Want. A real eigenvalue, i.e. a real root of m_T (Prop. 66).

Know. Over $\mathbb{R}$ there may be none: the rotation of $\mathbb{R}^2$ has $m = x^2 + 1$. But $\mathbb{R}^2$ has even dimension.

Plan. Show that "no real root" forces the dimension to be even, in three steps. (A) A real polynomial with no real root is a product of quadratics $x^2 + bx + c$ with $b^2 < 4c$. (B) For such a quadratic p, $\dim \operatorname{null} p(T)$ is even: each nonzero u in it comes with Tu, and $\operatorname{span}(u, Tu)$ is 2-dimensional. (C) Split off $\operatorname{null} p(T)$ and use induction on the dimension.

Step A: real polynomials. A real polynomial is a product of real linear factors and real quadratics $x^2 + bx + c$ with $b^2 < 4c$.

1. For real p, $p(\bar{z}) = \overline{p(z)}$, so if z is a non-real root, so is $\bar{z}$.
2. $q = (x-z)(x-\bar{z}) = x^2 - 2(\operatorname{Re} z)x + |z|^2$ is real, and $b^2 - 4c = -4(\operatorname{Im} z)^2 < 0$.
3. Divide over $\mathbb{R}$: $p = qs + r$ with $\deg r < 2$. Then $r(z) = r(\bar{z}) = 0$, and a polynomial of degree < 2 with two roots is 0. So $p = qs$ with s real. Real roots split off real linear factors (T6). Repeat on s (induction on the degree).

Step B: if $b^2 < 4c$ and $p = x^2 + bx + c$, then $\dim \operatorname{null} p(T)$ is even. Induction on $\dim \operatorname{null} p(T)$; let $U = \operatorname{null} p(T)$, which is invariant (T5).

1. $T|_U$ has no eigenvalue: $Tu = \lambda u$ with $u \in U$, $u \neq 0$, gives $0 = p(T)u = p(\lambda)u$, but $p(\lambda) = (\lambda + \frac{b}{2})^2 + c - \frac{b^2}{4} > 0$.
2. If $U = 0$, its dimension 0 is even. Otherwise pick $u \in U$, $u \neq 0$, and let $W = \operatorname{span}(u, Tu)$. It has dimension 2, since u is not an eigenvector. It is invariant, since $T(Tu) = -bTu - cu$.
3. The quotient operator $S = T|_U/W$ on U/W satisfies $p(S) = 0$ (T12), so $\operatorname{null} p(S)$ is all of U/W, of dimension $\dim U - 2$. By induction this is even, so $\dim U$ is even.

Step C: proof of F11. Induction on the odd number $n = \dim V$; $n = 1$ is clear.

1. Suppose T has no eigenvalue. Then m_T has no real root (Prop. 66), so by Step A, $m_T = ps$ with $p = x^2 + bx + c$, $b^2 < 4c$.
2. $s(T) \neq 0$ (smaller degree than m_T), so some w has $s(T)w \neq 0$, and $p(T)\big(s(T)w\big) = 0$. So $\operatorname{null} p(T) \neq 0$, and its dimension is even by Step B.
3. $W = \operatorname{range} p(T)$ is invariant, and $\dim W = n - \dim \operatorname{null} p(T)$ is odd and less than n (T1). $T|_W$ has no eigenvalue, because an eigenvector of $T|_W$ is one of T. This contradicts the induction hypothesis.

Computing m_T

Over $\mathbb{C}$, or whenever m_T splits into linear factors, $m_T = \prod_i (x - \lambda_i)^{k_i}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues (Prop. 66), every $k_i \geq 1$, and $\sum_i k_i \leq n$ (Prop. 67). So the work is finding the eigenvalues and then the exponents. Use whichever method below needs the least computation. (Ch. 14 adds a sharper bound on each k_i and one more method.)

Method 1: from the definition. Compute $I, A, A^2, \ldots$ and stop at the first power that is a combination of the earlier ones. Best for 2×2 matrices or when powers are easy.

Example

$A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ is not a multiple of I, so I, A are independent. $A^2 = \left(\begin{smallmatrix} 4 & 5 \\ 0 & 9 \end{smallmatrix}\right)$. Solve $A^2 = aA + bI$ entry by entry: entry $(1,2)$ gives $a = 5$; entry $(1,1)$ gives $4 = 2a + b$, so $b = -6$; entry $(2,2)$ checks: $9 = 3a + b$. So $m_A = x^2 - 5x + 6 = (x - 2)(x - 3)$.

Method 2: eigenvalues, then test candidates. Find the eigenvalues (for a triangular matrix, the diagonal entries, T8). The candidates are $\prod_i (x - \lambda_i)^{k_i}$ with every $k_i \geq 1$ and $\sum k_i \leq n$. Test them in increasing degree; the first one that kills A is m_A. Reason: m_A is one of the candidates, and any candidate q with $q(A) = 0$ is a multiple of m_A (Prop. 65), so $\deg q > \deg m_A$ unless $q = m_A$.

Example

$A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{smallmatrix}\right)$ has eigenvalues $3, 3, 1$. Candidates: degree 2: $(x - 3)(x - 1)$; degree 3: $(x - 3)^2(x - 1)$ and $(x - 3)(x - 1)^2$. $(A - 3I)(A - I) = \left(\begin{smallmatrix} 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right) \neq 0$, and $(A - 3I)^2(A - I) = 0$. So $m_A = (x - 3)^2(x - 1)$, and A is not diagonalizable.

$B = \left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right)$ has eigenvalues $1, 1, 2$, and $(B - I)(B - 2I) = 0$. So $m_B = (x - 1)(x - 2)$, and B is diagonalizable although it is not diagonal.

Method 3: blocks. For a block diagonal matrix, m is the lcm of the blocks' minimal polynomials (F4). For a block upper-triangular matrix $A = \left(\begin{smallmatrix} B & * \\ 0 & C \end{smallmatrix}\right)$: $m_B \mid m_A$, $m_C \mid m_A$ and $m_A \mid m_B m_C$. So if m_B and m_C have no common root, $m_A = m_B m_C$.

Why.

1. The first block of basis vectors spans an invariant subspace U with $B = \mathcal{M}(T|_U)$. C is the matrix of T/U in the cosets of the remaining basis vectors.
2. $m_B \mid m_A$ and $m_C \mid m_A$ by F3(a). $m_A \mid m_B m_C$ by step 4 of the proof of Prop. 67, which shows that $m_{T|_U} m_{T/U}$ kills T.

3. "No common root" means no common root in $\mathbb{C}$. Over $\mathbb{C}$, $m_A = \prod(x-\lambda)^{e_\lambda}$, and each monic divisor of it is $\prod(x-\lambda)^{b_\lambda}$ with $b_\lambda \le e_\lambda$ (its roots are roots of m_A; divide out one factor at a time, T6).
4. So m_B and m_C are such products over disjoint sets of λ's, and their product divides m_A.

- $A = \left(\begin{smallmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{smallmatrix}\right)$: the first block R has $R^2 = -I$ and is not a multiple of I, so $m_R = x^2+1$; the block $2I$ has $m = x-2$. So $m_A = (x-2)(x^2+1)$. Over $\mathbb{C}$ the roots $2, \pm i$ are distinct, so A is diagonalizable over $\mathbb{C}$; over $\mathbb{R}$ it is not, since $\pm i \notin \mathbb{R}$ (F8).
- The matrix A of Method 2 is $\left(\begin{smallmatrix} B & * \\ 0 & 1 \end{smallmatrix}\right)$ with $B = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, $m_B = (x-3)^2$. The roots 3 and 1 differ, so $m_A = (x-3)^2(x-1)$ with no matrix multiplication.

Method 4: follow one vector (F7). Compute $v, Av, A^2v, \ldots$ until the first dependence; it gives p_v, which divides m_A. If $\deg p_v = n$, then $m_A = p_v$. Otherwise repeat with more basis vectors and take the lcm.

Example

$A = \left(\begin{smallmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{smallmatrix}\right)$: $Ae_1 = e_2$, $Ae_2 = e_3$, so e_1, Ae_1, A^2e_1 is a basis. $A^3e_1 = Ae_3 = (2,-1,2) = 2e_1 - Ae_1 + 2A^2e_1$. So $m_A = x^3 - 2x^2 + x - 2 = (x-2)(x^2+1)$.

$A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{smallmatrix}\right)$: $Ae_1 = (2,1,0)$, $A^2e_1 = (4,4,0) = 4Ae_1 - 4e_1$, so $p_{e_1} = (x-2)^2$. $Ae_3 = 3e_3$, so $p_{e_3} = x-3$. The lcm $(x-2)^2(x-3)$ has degree $3 = n$, so it is m_A; there is no need to compute p_{e_2}.

The first matrix is a **companion matrix**: if $Ae_j = e_{j+1}$ for $j < n$ and $Ae_n = a_0e_1 + \cdots + a_{n-1}e_n$, then e_1 is cyclic and $A^ne_1 = Ae_n$, so $m_A = x^n - a_{n-1}x^{n-1} - \cdots - a_0$. So every monic polynomial is the minimal polynomial of some matrix.

Method 5: spot a relation. If you notice $p(A) = 0$, then $m_A \mid p$; check which divisors work.

- Rank one, $A = uv^{\mathsf{T}}$ (all columns multiples of u): $A^2 = u(v^{\mathsf{T}}u)v^{\mathsf{T}} = (\operatorname{tr} A)A$, where $\operatorname{tr} A$ is the sum of the diagonal entries, which equals $v^{\mathsf{T}}u$. For $n \ge 2$, A is not a multiple of I, so $m_A = x(x - \operatorname{tr} A)$ if $\operatorname{tr} A \ne 0$ (diagonalizable) and $m_A = x^2$ if $\operatorname{tr} A = 0$ (not diagonalizable). Example: $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)^2 = 2\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, so $m = x(x-2)$.
- $A^2 = A$: m_A is x, $x-1$ or $x(x-1)$. $A^2 = I$: m_A is $x-1$, $x+1$ or $(x-1)(x+1)$. Both cases are diagonalizable. $A^k = 0$: $m_A = x^j$ for the least j with $A^j = 0$.
- A permutation matrix satisfies $P^k = I$ for some $k \ge 1$ (only finitely many permutations, so two powers agree), so it is diagonalizable over $\mathbb{C}$ (F10).

Method 6: the 2×2 formula. For 2×2 $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, with $\operatorname{tr} A = a+d$ and $\det A = ad-bc$: $m_A = x^2 - (\operatorname{tr} A)x + \det A$, unless $A = cI$ (then $m_A = x-c$).

- Why: for $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $A^2 - (a+d)A = \left(\begin{smallmatrix} bc-ad & 0 \\ 0 & bc-ad \end{smallmatrix}\right)$, so $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \ne cI$, $\deg m_A = 2$ and m_A divides this monic quadratic, so they are equal.
- Example: $A = \left(\begin{smallmatrix} 1 & 4 \\ -1 & -3 \end{smallmatrix}\right)$: $\operatorname{tr} A = -2$, $\det A = 1$, so $m_A = (x+1)^2$; $A \ne -I$, so A is not diagonalizable.

Method 7: operators on function spaces. Work with the operator directly, or take its matrix in a basis.

Example

D on $V = \operatorname{span}(e^x, xe^x)$: $(D-I)e^x = 0$, $(D-I)(xe^x) = e^x \neq 0$, and $(D-I)^2(xe^x) = 0$. So $(D-I)^2 = 0$ and $D \neq I$: $m_D = (x-1)^2$, and D is not diagonalizable.

Shortcuts

- **No computation needed.** n distinct eigenvalues (e.g. a triangular matrix with distinct diagonal entries) $\Rightarrow m = \prod(x - \lambda_i)$ (every eigenvalue is a root and $\deg m \leq n$), diagonalizable. One eigenvalue λ and $A \neq \lambda I$ $\Rightarrow$ not diagonalizable.
- **Testing a candidate**: one nonzero entry is enough to show it fails.
- **Inverse** (F1). $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$: $A^2 - 5A + 6I = 0$, so $A(5I - A) = 6I$ and $A^{-1} = \frac{1}{6}(5I - A)$.
- **High powers.** Divide: $x^k = m(x)q(x) + r(x)$ with $\deg r < \deg m$; then $A^k = r(A)$. Find r by putting each root λ of m into this identity: $r(\lambda) = \lambda^k$. For a double root, differentiate as well: $r'(\lambda) = k\lambda^{k-1}$ (the derivative of $(x-\lambda)^2 q$ is $2(x-\lambda)q + (x-\lambda)^2 q'$, which vanishes at λ).
  - $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, $r = ax + b$: $2a + b = 2^k$, $3a + b = 3^k$, so $A^k = (3^k - 2^k)A + (3 \cdot 2^k - 2 \cdot 3^k)I$.
  - $m = (x+1)^2$ (as for $\left(\begin{smallmatrix} 1 & 4 \\ -1 & -3 \end{smallmatrix}\right)$): $r(-1) = (-1)^k$, $r'(-1) = k(-1)^{k-1}$, so $A^k = (-1)^k I + k(-1)^{k-1}(A + I)$.
- **Any polynomial** $p(A)$ equals $r(A)$, where r is the remainder of p divided by m_A. So every polynomial in A is a combination of $I, A, \ldots, A^{d-1}$.
- **Eigenvectors without solving** (Prop. 66). If $m = (x - \lambda)s$, the nonzero columns of $s(A)$ are eigenvectors for λ; if A is diagonalizable, they span $E(\lambda)$ (on $E(\mu)$, $s(A)$ is the number $s(\mu)$, which is 0 for $\mu \neq \lambda$ and nonzero for $\mu = \lambda$). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ and $\lambda = 3$: $s(A) = A - 2I = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, so $(1, 1)$ is an eigenvector for 3.
- **Diagonalizable without finding eigenvalues.** One polynomial that kills T and has distinct roots in $\mathbb{F}$ is enough, since m_T divides it: $T^2 = T$, $T^2 = I$, $T^3 = T$ (over $\mathbb{R}$), $T^k = I$ (over $\mathbb{C}$).
- **Not similar**: different minimal polynomials prove it (F5). Equal minimal polynomials do not prove similarity.
- **Lower bound**: if $v, Av, \ldots, A^{j-1}v$ are independent, then $\deg m_A \geq j$ (F7).

Traps

- Only roots in $\mathbb{F}$ are eigenvalues: over $\mathbb{R}$, $m = x^2 + 1$ gives no eigenvalues.
- A polynomial that kills T need not be m_T: $T^2 = T$ allows $m_T = x$, $x - 1$ or $x^2 - x$. Check the divisors.
- Every eigenvalue is a root of m_T, so a candidate that leaves out an eigenvalue never works.
- $p(T)v = 0$ for one $v \neq 0$ only gives $p_v \mid p$, not $m_T \mid p$. And v need not be an eigenvector even when p has distinct roots: $T = \operatorname{diag}(1, 2)$, $v = (1, 1)$, $p = (x-1)(x-2)$. What does follow, if p splits into linear factors over $\mathbb{F}$ (always true over $\mathbb{C}$), is that some root of p is an eigenvalue (T4, T6). Over $\mathbb{R}$ not even that: $T = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $p = x^2 + 1$, $v = e_1$.
- Over $\mathbb{R}$, $T^k = I$ does not imply diagonalizable: $R = \left(\begin{smallmatrix} 1 & -2 \\ 1 & -1 \end{smallmatrix}\right)$ has $R^4 = I$ but $m_R = x^2 + 1$.
- $m_{T^{-1}}(x)$ is not $m_T(1/x)$, which is not a polynomial. Reverse the coefficients and divide by c_0.
- $\deg m_T = n$ says nothing about diagonalizability: $\operatorname{diag}(1, 2)$ and $J_2(0)$ both have degree 2.

§1.3 Generalized eigenvectors (Ch. 14)

The question. We want the simplest matrix for every operator. With an eigenbasis it is diagonal, but often there are too few eigenvectors.

- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has eigenvectors only on $\operatorname{span}(e_1)$.
- But $Ne_2 = e_1 \neq 0$ and $N^2 e_2 = 0$: e_2 is sent to 0 by N^2 but not by N.
- Idea of the chapter: allow $(T - \lambda I)^k v = 0$ for some $k \geq 1$ instead of only $k = 1$. Over $\mathbb{C}$ that gives enough vectors for a basis.

Definition (Def. 37 / LADR 8.8, 8.19)**.** $v \neq 0$ is a **generalized eigenvector** for λ if $(T - \lambda I)^k v = 0$ for some $k \geq 1$. The **generalized eigenspace** is $G(\lambda, T) = \bigcup_{k \geq 0} \operatorname{null}(T - \lambda I)^k$.

- $E(\lambda, T) \subseteq G(\lambda, T)$ (take $k = 1$). In the example, $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$.
- A union over all k is awkward. Props. 69–70 show it stops at a fixed power.
- An operator N is **nilpotent** if $N^k = 0$ for some k (Def. 39), e.g. N above. So $G(\lambda, T)$ is the part of V on which $T - \lambda I$ acts nilpotently.

$$\underset{\text{generalized eigenvector}}{v} \xrightarrow{T-\lambda I} (T-\lambda I)v \xrightarrow{T-\lambda I} \cdots \xrightarrow{T-\lambda I} \underset{\text{last nonzero vector: an eigenvector}}{w} \xrightarrow{T-\lambda I} 0$$

Repeatedly applying $T - \lambda I$ to a generalized eigenvector reaches 0. The last nonzero vector w has $(T - \lambda I)w = 0$, so it is an eigenvector (Prop. 71). Smallest case: $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_2 \mapsto e_1 \mapsto 0$.

Proposition (Prop. 69 / LADR 8.1, 8.2: the null-space chain)
$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T \subseteq \operatorname{null} T^2 \subseteq \cdots$. If $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^m = \operatorname{null} T^{m+k}$ for all $k \geq 0$.

Where the proof comes from

Want. (a) Each null space contains the one before. (b) Once two neighbours are equal, all later ones are equal.
(a) A vector killed by T^k stays killed by $T^{k+1} = T \cdot T^k$.
Know, for (b). Only one equality, $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. To use it on a vector $v \in \operatorname{null} T^{m+k+1}$, we need some vector in $\operatorname{null} T^{m+1}$ built from v.
Key step. Split the power: $T^{m+k+1} v = T^{m+1}(T^k v)$. So $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$, which says $T^{m+k} v = 0$. That is the equality one step higher; repeat.
Example. The shift on $\mathbb{F}^3$ ($e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$) has $\dim \operatorname{null} T^k = 0, 1, 2, 3, 3, 3, \ldots$: it grows until two neighbours are equal, and after that all are equal.

Proof. 1. If $T^k v = 0$, then $T^{k+1} v = T(0) = 0$. So $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$.
2. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $T^{m+k+1} v = 0$. Then $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$.
3. So $T^{m+k} v = T^m(T^k v) = 0$. Hence $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, and with step 1 they are equal. Induct on k.

□

Proposition (Prop. 70 / LADR 8.3, 8.20)
$\operatorname{null} T^n = \operatorname{null} T^{n+1}$. Hence $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, a T-invariant subspace.

Where the proof comes from

Want. The chain has stopped by the power n.

Know. Until it stops, each step raises the dimension by at least 1 (a strictly larger subspace has larger dimension). Dimensions are at most n. Once it stops, it stays stopped (Prop. 69).

Key step. Count: starting from dimension 0, $n+1$ strict steps would reach dimension $n+1 > n$. So one of the first $n+1$ steps is an equality, and from there on the chain is constant.

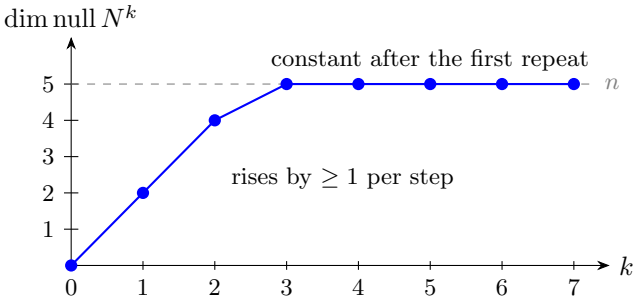
Why it matters. The union defining $G(\lambda, T)$ is then the single null space $\operatorname{null}(T-\lambda I)^n$. So it is a subspace, and it is invariant by (T5).

Proof. 1. Suppose $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$. By Prop. 69 no earlier pair is equal either.

2. Then $0 = \dim \operatorname{null} T^0 < \dim \operatorname{null} T^1 < \cdots < \dim \operatorname{null} T^{n+1} \leq n$: that is $n+2$ distinct integers in $\{0, \ldots, n\}$. Impossible.

3. So $\bigcup_k \operatorname{null}(T-\lambda I)^k = \operatorname{null}(T-\lambda I)^n$. It is invariant by (T5).

□



N with blocks $J_3(0), J_2(0)$ on $\mathbb{F}^5$: $\dim \operatorname{null} N^k = 0, 2, 4, 5, 5, \ldots$ While the chain grows it gains at least 1 per step, and it cannot pass n, so it has stopped by $k = n$ (Prop. 70). Once two neighbours agree, all later ones agree (Prop. 69).

Proposition (LADR 8.4, 8.5: splitting off the nilpotent part)

$V = \operatorname{null} T^n \oplus \operatorname{range} T^n$, and T is injective on $\operatorname{range} T^n$.

Where the proof comes from

Want. Split V into a part where T is nilpotent ($\operatorname{null} T^n$) and a part where T is invertible ($\operatorname{range} T^n$). Applied to $T-\lambda I$, this splits off $G(\lambda, T)$; that is the first step of Prop. 73.

Know. $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1). So by (T3) it is enough to show $\operatorname{null} T^n \cap \operatorname{range} T^n = \{0\}$.

First try. Power 1: is $\operatorname{null} T \cap \operatorname{range} T = \{0\}$? No. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_1 = Ne_2$ is in the range and $Ne_1 = 0$. A nonzero vector of range N can lie in null N.

Key step. With the power n, $\operatorname{range} T^n$ meets $\operatorname{null} T^n$ only in 0, because the chain has stopped. If $v = T^n u$ and $T^n v = 0$, then $T^{2n} u = 0$, and $\operatorname{null} T^{2n} = \operatorname{null} T^n$ (Prop. 70) gives $T^n u = 0$, i.e. $v = 0$.

Proof. 1. Let $v = T^n u$ with $T^n v = 0$. Then $T^{2n} u = 0$, so $u \in \operatorname{null} T^{2n} = \operatorname{null} T^n$, so $v = 0$.

2. $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1), so $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (T3).

3. If $w \in \operatorname{range} T^n$ and $Tw = 0$, then $w \in \operatorname{null} T \subseteq \operatorname{null} T^n$, so $w = 0$.

□

Proposition (Prop. 71)

$G(\lambda, T) \neq \{0\} \iff \lambda$ is an eigenvalue of T.

Where the proof comes from

Want. A nonzero generalized eigenvector for λ forces an eigenvector for λ.

Key step. Apply $T - \lambda I$ to v repeatedly: $v \mapsto (T-\lambda I)v \mapsto \cdots \mapsto 0$. The last nonzero vector in this list is sent to 0 by $T - \lambda I$, so it is an eigenvector.

Proof. If λ is an eigenvalue, $\{0\} \neq E(\lambda, T) \subseteq G(\lambda, T)$. Conversely, let $v \in G(\lambda, T)$, $v \neq 0$, and let $j \geq 0$ be the largest exponent with $w = (T-\lambda I)^j v \neq 0$ (it exists because $(T-\lambda I)^k v = 0$ for some k). Then $(T-\lambda I)w = 0$ and $w \neq 0$. □

Proposition (Prop. 72 / LADR 8.12)

Generalized eigenvectors $v_1, \ldots, v_m$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ are independent.

Where the proof comes from

Want. From $\sum a_i v_i = 0$, get each $a_k = 0$.

Know. How independence of eigenvectors is proved (Prop. 54): apply an operator that kills every term except the k-th and is nonzero on the k-th. What is left is $a_k \cdot (\text{nonzero vector}) = 0$.

First try. $(T-\lambda_i I)^n$ kills v_i, so $S = \prod_{i \neq k}(T-\lambda_i I)^n$ kills every v_i with $i \neq k$. But it is not clear what S does to v_k, which is only a generalized eigenvector.

Key step. On an actual eigenvector w for λ_k, S is easy: $Sw = \prod_{i \neq k}(\lambda_k - \lambda_i)^n w \neq 0$ (T7). So first turn v_k into an eigenvector by applying $(T-\lambda_k I)^N$, as in Prop. 71. The extra factor still gives $Sv_i = 0$ for $i \neq k$, because all the factors commute.

Proof. 1. Fix k. Let N be the largest exponent with $w = (T-\lambda_k I)^N v_k \neq 0$. Then $Tw = \lambda_k w$, so $(T-\lambda_i I)^n w = (\lambda_k - \lambda_i)^n w$.

2. Apply $S = \prod_{i \neq k}(T-\lambda_i I)^n \cdot (T-\lambda_k I)^N$ to $\sum a_i v_i = 0$. The factors commute (T5) and $(T-\lambda_i I)^n v_i = 0$ (Prop. 70), so $Sv_i = 0$ for $i \neq k$.

3. What is left: $0 = a_k \prod_{i \neq k}(\lambda_k - \lambda_i)^n \, w$. The product and w are nonzero, so $a_k = 0$.

□

Two consequences.

- $G(\lambda, T) \cap G(\mu, T) = \{0\}$ for $\lambda \neq \mu$: a nonzero v in both would make the list v, v independent (Prop. 72). So a generalized eigenvector belongs to only one eigenvalue.
- If U is invariant, $G(\lambda, T|_U) = G(\lambda, T) \cap U$, since $(T|_U - \lambda I)^k u = (T - \lambda I)^k u$ for $u \in U$.

Theorem (Prop. 73 / LADR 8.9, 8.22: generalized eigenspace decomposition)
Over $\mathbb{C}$, if $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, then $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. In particular V has a basis of generalized eigenvectors of T.

Where the proof comes from

Want. (a) The $G(\lambda_i, T)$ together span V. (b) The sum is direct.

(b) is Prop. 72: a relation $u_1 + \cdots + u_m = 0$ with $u_i \in G(\lambda_i, T)$ and some $u_i \neq 0$ would be a dependence among generalized eigenvectors for distinct eigenvalues.

Know, for (a). The splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$ with $U = \operatorname{range}(T - \lambda_1 I)^n$ invariant. U is smaller than V because $G(\lambda_1, T) \neq 0$.

Key step. Induct on $\dim V$. The theorem for $T|_U$ writes U as a sum of generalized eigenspaces of $T|_U$, and these lie inside those of T. The eigenvalue λ_1 does not come back in U, because $T - \lambda_1 I$ is injective on U.

Why $\mathbb{C}$. We need an eigenvalue λ_1 to start (T7). Over $\mathbb{R}$, the rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has no eigenvalues, so all its generalized eigenspaces are 0.

Example. $A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$: $G(1, A) = \operatorname{span}(e_1, e_2)$ (Ex. 61 below) and $G(3, A) = E(3, A) = \operatorname{span}\big((3, 2, 4)\big)$. So $\mathbb{C}^3 = G(1, A) \oplus G(3, A)$.

Proof. 1. Induction on n; $n = 1$ is clear. Take an eigenvalue λ_1 (T7).

2. Splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$, $U = \operatorname{range}(T - \lambda_1 I)^n$. U is invariant (T5), and $\dim U < n$ since $G(\lambda_1, T) \neq 0$ (Prop. 71).
3. Eigenvalues of $T|_U$ are eigenvalues of T, but not λ_1 ($T - \lambda_1 I$ is injective on U). By induction $U = \bigoplus_{i \geq 2} G(\lambda_i, T|_U)$, and $G(\lambda_i, T|_U) \subseteq G(\lambda_i, T)$.
4. So the $G(\lambda_i, T)$ span V. Direct: if $u_1 + \cdots + u_m = 0$, $u_i \in G(\lambda_i, T)$, the nonzero u_i would be independent (Prop. 72); so all are 0.

□

On each $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent. So everything reduces to nilpotent operators (Ch. 15).

Definition (Def. 38). **Algebraic multiplicity** of λ: $\dim G(\lambda, T)$. **Geometric multiplicity**: $\dim E(\lambda, T)$.

- geometric $\leq$ algebraic, since $E \subseteq G$.
- Over $\mathbb{C}$, the algebraic multiplicities add up to n (Prop. 73).
- Over $\mathbb{C}$, T is diagonalizable $\iff$ geometric = algebraic for every eigenvalue.
  Why. The sum $\sum E(\lambda_i, T)$ is direct (T7), and T is diagonalizable iff it is V, i.e. $\sum \dim E = n = \sum \dim G$. Since each $\dim E \leq \dim G$, the sums agree iff every term agrees.

Theorem (Prop. 74 / LADR 8.26, 8.29: Cayley–Hamilton)
Over $\mathbb{C}$, let $d_i = \dim G(\lambda_i, T)$ and $\chi_T(x) = \prod_i (x - \lambda_i)^{d_i}$ (the **characteristic polynomial**). Then $\chi_T(T) = 0$, so $m_T \mid \chi_T$. Also $\deg \chi_T = \sum d_i = n$ (Prop. 73), so $\deg m_T \leq n$.

Where the proof comes from

Want. $\chi_T(T)v = 0$ for every v.
Know. $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73), so it is enough to take v in one $G(\lambda_i, T)$. There, $T - \lambda_i I$ is nilpotent.
Key step 1. Which power of $T - \lambda_i I$ kills $G(\lambda_i, T)$? Prop. 70, applied inside $G(\lambda_i, T)$, a space of dimension d_i, gives the power d_i. That is why the exponents in χ_T are the d_i.
Key step 2. The factors of $\chi_T(T)$ commute, so for $v \in G(\lambda_i, T)$ let $(T - \lambda_i I)^{d_i}$ act first.

Proof. 1. $N = (T - \lambda_i I)|_{G(\lambda_i, T)}$ has $G(0, N) = G(\lambda_i, T)$, so $N^{d_i} = 0$ by Prop. 70 in that space.
2. For $v \in G(\lambda_i, T)$: $\chi_T(T)v = \prod_{j \neq i}(T - \lambda_j I)^{d_j}\,(T - \lambda_i I)^{d_i} v = 0$ (T5).
3. The G's span V, so $\chi_T(T) = 0$; then $m_T \mid \chi_T$ (Prop. 65) and $\deg \chi_T = \sum d_i = n$.

□

- For a matrix, $\chi_A(x) = \det(xI - A)$ (stated in the notes; needs determinants).
- 2×2: $\chi_A = x^2 - (\operatorname{tr} A)x + \det A$.
  Why. Multiplying out gives $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq cI$, m_A has degree 2 and divides both this polynomial and χ_A, so all three agree. If $A = cI$, both are $(x - c)^2$.
- χ_T and m_T have the same roots; the exponents in m_T can be smaller.

Proposition (Exponents of m_T over $\mathbb{C}$, M6)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and $d_i = \dim G(\lambda_i, T)$. Then $m_T = \prod_i (x - \lambda_i)^{s_i}$, where

$$s_i = \text{the least } k \text{ with } \operatorname{null}(T - \lambda_i I)^k = G(\lambda_i, T), \qquad 1 \leq s_i \leq d_i.$$

In particular $m_T \mid \chi_T$, and m_T has the same roots as χ_T.

Where the proof comes from

Want. The exponent of each $x - \lambda_i$ in m_T.
Know. V splits into the invariant pieces $G(\lambda_i, T)$ (Prop. 73), and the minimal polynomial of a direct sum is the lcm of the pieces' minimal polynomials (F4).
Key step. So compute m on each piece separately. On $G(\lambda_i, T)$ the operator $T - \lambda_i I$ is nilpotent, so the minimal polynomial there is a power of $x - \lambda_i$, and the right power is the first one that is 0 on the piece.

Proof. 1. $(T - \lambda_i I)^n = 0$ on $G(\lambda_i, T)$ (Prop. 70), so $m_{T|_{G(\lambda_i, T)}}$ divides $(x - \lambda_i)^n$. Its roots are roots of $(x - \lambda_i)^n$, so it is $(x - \lambda_i)^{s_i}$ for some s_i (over $\mathbb{C}$ it splits into linear factors, T6).
2. s_i is the least k with $(T - \lambda_i I)^k = 0$ on $G(\lambda_i, T)$, i.e. with $\operatorname{null}(T - \lambda_i I)^k \supseteq G(\lambda_i, T)$. Since $\operatorname{null}(T - \lambda_i I)^k \subseteq G(\lambda_i, T)$ always, this means equality.
3. By Prop. 73 and F4, $m_T = \operatorname{lcm}_i (x - \lambda_i)^{s_i} = \prod_i (x - \lambda_i)^{s_i}$ (the λ_i are distinct, so nothing is shared).
4. $s_i \geq 1$ since $G(\lambda_i, T) \neq 0$, and $s_i \leq d_i$ by Prop. 70 applied inside the d_i-dimensional $G(\lambda_i, T)$.

□

Computing m_T from null-space dimensions. For each eigenvalue λ, compute $d_k = \dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$ for $k = 1, 2, \ldots$. The first k at which d_k stops growing (equivalently, reaches the algebraic multiplicity) is the exponent of $x - \lambda$.

Example
$A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$ (Ex. 61). For $\lambda = 1$: $d_1 = 1$ and $d_2 = 2$, which is the number of 1s on the diagonal, so the exponent is 2. $\lambda = 3$ appears once, so its exponent is 1. So $m_A = (x - 1)^2(x - 3)$.

- With the multiplicities d_i known, Method 2 of Ch. 13 needs only the candidates with $k_i \le d_i$, and χ_T itself never needs testing (Cayley–Hamilton). In the Method 2 example, 1 has multiplicity 1, so $(x-3)(x-1)^2$ was never a candidate, and $(x-3)^2(x-1) = \chi_A$ needed no check.
- For 3×3 A (using $\chi_A = \det(xI-A)$, stated in the notes; determinants are Ch. 25): $\chi_A = x^3-(\operatorname{tr} A)x^2+c_2x-\det A$, where c_2 is the sum of the three 2×2 minors on the diagonal (rows and columns $\{1,2\},\{1,3\},\{2,3\}$). This gives the eigenvalues and multiplicities for Method 2 when A is not triangular.

Proposition (LADR 8.31: diagonal of a triangular matrix)
Over $\mathbb{C}$, if $\mathcal{M}(T)$ is upper-triangular, then $\dim G(\lambda, T)$ is the number of times λ appears on the diagonal.

Where the proof comes from

Want. $\dim G(\lambda,T) = c_\lambda$, the number of times λ appears on the diagonal.

Know. $\dim G(\lambda,T) = \dim \operatorname{null}(T-\lambda I)^n = n - \operatorname{rank}(T-\lambda I)^n$. The matrix of $(T-\lambda I)^n$ is upper-triangular with diagonal entries $(a_{jj}-\lambda)^n$, nonzero in exactly $n-c_\lambda$ places.

First try. Compute the rank exactly. This is hard: columns with a 0 on the diagonal can still be nonzero and add to the rank.

Key step. Settle for an inequality. The $n-c_\lambda$ columns with nonzero diagonal entries are independent (T8), so rank $\ge n-c_\lambda$, i.e. $\dim G(\lambda,T) \le c_\lambda$. Now add over all λ: both sides add up to n, so none of the inequalities can be strict.

Pattern. "Prove $\le$ for each term, then compare totals" also proves the test "diagonalizable $\iff$ geometric $=$ algebraic" above.

Proof. 1. Let c_λ be the count. $\mathcal{M}((T-\lambda I)^n)$ is triangular with diagonal $(a_{jj}-\lambda)^n$, nonzero in $n-c_\lambda$ places.
2. Those columns are independent (T8), so rank $\ge n-c_\lambda$ and $\dim G(\lambda,T) \le c_\lambda$ (Prop. 70, T1).
3. $n = \sum_\lambda \dim G(\lambda,T) \le \sum_\lambda c_\lambda = n$ (Prop. 73), so every inequality is an equality.

□

Example (Computing $G(\lambda, A)$ (Ex. 61))
$A = \left(\begin{smallmatrix}1&1&1\\0&1&1\\0&0&3\end{smallmatrix}\right)$, $\lambda = 1$.

1. $\operatorname{null}(A-I) = \operatorname{span}(e_1)$, $\operatorname{null}(A-I)^2 = \operatorname{span}(e_1,e_2)$, $\operatorname{null}(A-I)^3 = \operatorname{span}(e_1,e_2)$.
2. Two consecutive null spaces agree, so the chain has stopped: $G(1,A) = \operatorname{span}(e_1,e_2)$.
3. Shortcut: 1 appears twice on the diagonal, so you can stop at dimension 2.

Shortcuts

- **Exponent 1 test.** The exponent of λ in m is 1 $\iff$ $\dim E(\lambda)$ equals the algebraic multiplicity $\iff$ $\operatorname{rank}(A-\lambda I) = n - d_\lambda$. One rank computation per eigenvalue settles diagonalizability.
- **Never test χ itself** when computing m: it always works (Cayley–Hamilton). If every smaller candidate fails, $m = \chi$.
- **Smaller power.** $G(\lambda, T) = \operatorname{null}(T-\lambda I)^d$, d the algebraic multiplicity, and even $\operatorname{null}(T-\lambda I)^k$ with k the exponent of $x-\lambda$ in m_T (exponents of m_T, above). Why: $T-\lambda I$ is nilpotent on the d-dimensional space $G(\lambda, T)$, so Prop. 70 inside it gives the power d.
- **Know the target dimension first.** For a triangular matrix, $\dim G(\lambda)$ is the number of λ's on the diagonal (LADR 8.31); in general it is the multiplicity of λ as a root of $\chi_A = \det(xI - A)$. Stop the chain $\dim \operatorname{null}(A-\lambda I)^k$ as soon as it reaches that number.
- **Last one by subtraction.** The algebraic multiplicities add up to n, so the last one is n minus the others. And if the $\dim G(\lambda_i)$ you found already add up to n, there are no other eigenvalues.
- **Multiplicity 1.** If the algebraic multiplicity is 1, then $G(\lambda) = E(\lambda)$ (since $1 \le \dim E \le \dim G = 1$).
- **One eigenvalue.** Over $\mathbb{C}$, if λ is the only eigenvalue, $G(\lambda, T) = V$ and $(T-\lambda I)^n = 0$ with no computation.
- **Geometric multiplicity by rank.** $\dim E(\lambda) = n - \operatorname{rank}(A-\lambda I)$, and $\dim \operatorname{null}(A-\lambda I)^k = n - \operatorname{rank}(A-\lambda I)^k$.
- $G(\lambda)$ **as a range.** Over $\mathbb{C}$, $G(\lambda, T) = \operatorname{range} \prod_{\mu \neq \lambda} (T-\mu I)^{d_\mu}$, and $\bigoplus_{\mu \neq \lambda} G(\mu, T)$ is the null space of that product. Useful when the other eigenvalues are simple: for the A of Ex. 61, $G(1, A) = \operatorname{range}(A - 3I) = \operatorname{span}(e_1, e_2)$. Why, with P the product and $H = \bigoplus_{\mu \neq \lambda} G(\mu, T)$:
  1. $P = 0$ on each $G(\mu, T)$, $\mu \neq \lambda$: let $(T-\mu I)^{d_\mu}$ act first (Prop. 74, step 1).
  2. P is injective on $G(\lambda, T)$: each $T - \mu I$ is, since a nonzero $v \in G(\lambda, T)$ with $Tv = \mu v$ would lie in $G(\lambda, T) \cap G(\mu, T) = 0$.
  3. $G(\lambda, T)$ is invariant (T5), so P maps it injectively into itself, hence onto itself (T2).
  4. $V = G(\lambda, T) \oplus H$ (Prop. 73) and $P = 0$ on H, so $\operatorname{range} P = G(\lambda, T)$ and $\operatorname{null} P = H$.
- **Commuting operators.** If $ST = TS$, then S maps $G(\lambda, T)$ into itself (G2): $(T-\lambda I)^n Sv = S(T-\lambda I)^n v = 0$.

Traps

- $G(\lambda, T)$ can be strictly bigger than $E(\lambda, T)$. Over $\mathbb{C}$ they agree for all λ exactly when T is diagonalizable.
- The exponents in m_T are at most the algebraic multiplicities, and they are not the geometric multiplicities: on $\mathbb{F}^4$ with $Ne_2 = e_1$ and $Ne_1 = Ne_3 = Ne_4 = 0$, $m_N = x^2$ while $\dim E(0) = 3$ and $\dim G(0) = 4$.
- A sum of generalized eigenvectors for different eigenvalues is not a generalized eigenvector (Prop. 72 makes the pieces independent).
- Prop. 73 needs eigenvalues: true over $\mathbb{C}$; over $\mathbb{R}$ only when m_T splits. (If m_T splits, every $T|_U$ with $U \neq 0$ invariant has an eigenvalue, because $m_{T|_U} \mid m_T$ (F3) and its roots are eigenvalues (Prop. 66); so the proof of Prop. 73 still works.)
- $V = \operatorname{null} T \oplus \operatorname{range} T$ can fail: for $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$. The splitting needs the power n.
- The power n always suffices; the chain often stops sooner.

§1.4 Nilpotent operators (LADR 8A, 8B)

Definition (Def. 39)**.** N is **nilpotent** if $N^k = 0$ for some k; the least such k is the **nilpotency index**.

- $N^n = 0$: $V = G(0, N) = \operatorname{null} N^n$ (Prop. 70). So $m_N = x^k$, $k \le n$.
- The only eigenvalue is 0 (if $V \neq 0$): $Nv = \lambda v \Rightarrow 0 = N^k v = \lambda^k v$. Over $\mathbb{C}$, conversely, only eigenvalue $0 \Rightarrow V = G(0, N)$ (Prop. 73).
- $N \neq 0$ is not diagonalizable: a diagonal matrix with all eigenvalues 0 is 0.
- Index $= \deg m_N \le n$. HW8 P1: D on $\mathcal{P}_n(\mathbb{C})$ has index $n+1$, since $D^n x^n = n! \neq 0$ and $D^{n+1} = 0$.
- $(I - N)^{-1} = I + N + \cdots + N^{n-1}$: the product telescopes to $I - N^n = I$.

Proposition (LADR 8.18: nilpotent operators are strictly upper-triangular)
If N is nilpotent, then V has a basis in which $\mathcal{M}(N)$ is upper-triangular with every diagonal entry 0.

Where the proof comes from

Want. A basis $v_1, \ldots, v_n$ with $Nv_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ for every j. That is exactly an upper-triangular matrix with zero diagonal.

Know. If $N^k v = 0$, then $N^{k-1}(Nv) = 0$. So N maps $\operatorname{null} N^k$ into $\operatorname{null} N^{k-1}$.

Key step. List vectors by the least power of N that sends them to 0: a basis of $\operatorname{null} N$, extended to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$. A vector added at stage k is sent into $\operatorname{null} N^{k-1}$, which is spanned by vectors listed earlier.

Example. $N = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ ($Ne_1 = e_2$, $Ne_2 = 0$): $\operatorname{null} N = \operatorname{span}(e_2)$, so list e_2 first. In the basis e_2, e_1 the matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

Proof. 1. Choose a basis of $\operatorname{null} N$, extend it to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$ (Props. 69, 70).
2. If v was added for $\operatorname{null} N^k$, then $N^{k-1}(Nv) = N^k v = 0$, so $Nv \in \operatorname{null} N^{k-1}$, the span of the earlier layers (for $k = 1$, $Nv = 0$).
3. So each basis vector is sent into the span of the vectors listed before it: column j has nonzero entries only above the diagonal.

□

Theorem (LADR 8.37: block diagonal form)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T and $d_i = \dim G(\lambda_i, T)$. There is a basis in which $\mathcal{M}(T) = \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix}$, where each A_i is a $d_i \times d_i$ upper-triangular matrix with every diagonal entry equal to λ_i.

Where the proof comes from

Want. A block diagonal matrix whose i-th block is upper-triangular with every diagonal entry λ_i.

Know. Prop. 73 splits V into invariant pieces $G(\lambda_i, T)$, and invariant pieces give a block diagonal matrix (T10). So only the basis inside each piece is left to choose.

Key step. On $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent, and $\lambda_i I$ has the matrix $\lambda_i I$ in every basis. So choose the basis that makes N_i strictly upper-triangular (LADR 8.18); adding $\lambda_i I$ puts λ_i on the diagonal.

Proof. 1. $N_i = (T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent, since $G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). By LADR 8.18, $G(\lambda_i, T)$ has a basis in which $\mathcal{M}(N_i)$ is upper-triangular with zero diagonal.
2. In that basis $\mathcal{M}(T|_{G(\lambda_i, T)}) = \lambda_i I + \mathcal{M}(N_i)$: upper-triangular, diagonal entries λ_i, size d_i.
3. $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73) with each piece invariant, so concatenating the bases gives a block diagonal matrix (T10).

□

$$\mathcal{M}(T) = \left(\begin{array}{ccc|cc} \lambda_1 & * & * & 0 & 0 \\ 0 & \lambda_1 & * & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ \hline 0 & 0 & 0 & \lambda_2 & * \\ 0 & 0 & 0 & 0 & \lambda_2 \end{array}\right)$$

columns 1–3: a basis of $G(\lambda_1, T)$, $d_1 = 3$
columns 4–5: a basis of $G(\lambda_2, T)$, $d_2 = 2$

The zero blocks come from splitting V into invariant pieces (Prop. 73, T10). Each diagonal block is $\lambda_i I$ plus a strictly upper-triangular nilpotent part (LADR 8.18).

§2 Problem set

§2.1 Minimal polynomial (HW7)

HW7 P1. Return to the matrices from Homework 6, $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(a) Find the minimal polynomial of A over $\mathbb{R}$ and over $\mathbb{C}$.
(b) Find the minimal polynomial of B over $\mathbb{R}$.
(c) Use the minimal-polynomial criterion for diagonalizability to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

HW7 P2. Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation.

(a) Find the minimal polynomial of D.
(b) Decide whether D is diagonalizable.

HW7 P3. Let T be an invertible operator on a finite-dimensional vector space, and let $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ be its minimal polynomial.

(a) Prove that $a_0 \neq 0$.
(b) Prove that T^{-1} is a polynomial in T.
(c) Suppose now that V is a complex vector space and that $m_T(x) = x^4 + x^3 - x^2 - 5x + 4$. Express T^{-1} explicitly as a polynomial in T, find the eigenvalues of T, and decide whether T is diagonalizable.

HW7 P4. (optional) Let m_T be the minimal polynomial of an operator T.

(a) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$.
(b) If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of cT is $c^d m_T(x/c)$.

HW7 P5. Let A be a 2×2 complex matrix whose only eigenvalue is λ. Prove that its minimal polynomial is either $x - \lambda$ or $(x - \lambda)^2$. Show that the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable if and only if $A = \lambda I$.

HW7 P6. Consider $C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether C is diagonalizable.

HW7 P7. (optional) Let T be an operator whose minimal polynomial splits as $m_T(x) = (x - \lambda_1)\cdots(x - \lambda_r)$ with distinct roots. For a fixed j, construct a polynomial p_j such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

M1. Computing minimal polynomials. Find the minimal polynomial and decide whether the matrix is diagonalizable.

(a) $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

(b) $P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, over $\mathbb{R}$ and over $\mathbb{C}$.

(c) $B = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}$.

M2. Operators on polynomials. Let $n \geq 1$.

(a) Let $T\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $T(p) = xp'$. Find m_T and decide whether T is diagonalizable.
(b) Let $S\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $(Sp)(x) = p(x+1)$. Prove that $m_S = (x-1)^{n+1}$, and decide whether S is diagonalizable.

M3. Let V be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$ with $T^3 = T$. Prove that T is diagonalizable and that every eigenvalue of T is -1, 0 or 1.

M4. Let $V \neq \{0\}$ be finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $\deg m_T = 1$ if and only if $T = cI$ for some $c \in \mathbb{F}$.

M5. Minimal polynomial of the inverse. Let T be invertible with $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Prove that $m_{T^{-1}}(x) = \frac{1}{a_0}x^d m_T(1/x)$.

M6. Minimal polynomial and generalized eigenspaces. Let V be a finite-dimensional complex vector space and $m_T = (x-\lambda_1)^{k_1}\cdots(x-\lambda_m)^{k_m}$ with distinct λ_i. Prove that k_i is the smallest k with $\operatorname{null}(T-\lambda_i I)^k = G(\lambda_i, T)$.

M9. Using the minimal polynomial to compute. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Find m_A, write A^{-1} as a polynomial in A, and write A^3 as $aA + bI$.

§2.2 Generalized eigenvectors (HW8)

HW8 P1. Let $D\colon \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$ be differentiation.

(a) Prove that D is nilpotent and determine its nilpotency index.
(b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.

HW8 P3. Let $J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Compute $\operatorname{null}(J-I)$, $\operatorname{null}((J-I)^2)$, and $\operatorname{null}((J-I)^3)$.
(b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.
(c) Compute J^m for every positive integer m.

G1. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ act on $\mathbb{C}^3$. Find the eigenvalues, $E(2, A)$, $G(2, A)$, and m_A.

G2. Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$ with $ST = TS$. Prove that $G(\lambda, T)$ is invariant under S.

G3.

(a) Let N be nilpotent on a finite-dimensional $V \neq \{0\}$. Prove that $\lambda I + N$ is invertible if and only if $\lambda \neq 0$.
(b) Let V be a finite-dimensional complex vector space and suppose $T \in \mathcal{L}(V)$ has only one eigenvalue λ. Prove that $T - \lambda I$ is nilpotent.

G5. Let V be a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that if every generalized eigenvector of T is an eigenvector, then T is diagonalizable.